

TARAN MAMIDALA

SOFTWARE ENGINEER | DATA ENGINEER | AI/ML

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SUMMARY

Software, data, and AI engineer with 3+ years of non-overlapping experience across backend development, data pipelines, applied research, and production-facing software. M.S. in Computer Science and Engineering with an AI/ML specialization. Processed 2M+ satellite records and contributed to peer-reviewed research.

EXPERIENCE

Rebecca Everlene Trust Company

Software / Data Analytics Intern

- Support structured data workflows, reporting, validation, and process automation for a mission-driven nonprofit.
- Apply Python and related tools to organize data, document workflows, and improve repeatable operational processes.
- Collaborate asynchronously while following confidentiality, communication, and data-handling requirements.

Mar 2026 - Present

Remote, United States

University at Buffalo, SUNY

Research Assistant

- Built reproducible Python workflows for data preparation, analysis, and machine learning experiments.
- Documented methods, assumptions, and results so research work could be reviewed, repeated, and extended.
- Created analysis tooling and visual explanations for technical and non-technical research audiences.

Jan 2025 - Jan 2026

Buffalo, NY

iVinGo Solutions Pvt. Ltd.

Software Engineer, Backend & Data

- Delivered web platforms, backend services, database-backed features, reporting workflows, QA, debugging, and documentation for client projects.
- Worked with Python, Java, JavaScript, REST APIs, SQL, MySQL, Oracle, PostgreSQL, and MongoDB across education, pharma, travel, e-commerce, and marketing use cases.
- Supported data ingestion, cleaning, validation, schema updates, query development, Docker workflows, and CI/CD with GitHub Actions and Jenkins.

Jun 2022 - Jul 2024

Hyderabad, India / Remote

NRSC, Indian Space Research Organisation (ISRO)

Research Intern, Data Science & Geospatial AI (Concurrent)

- Processed and validated 2M+ satellite records for atmospheric CO2, climate-zone, and land-use/land-cover analysis.
- Built reproducible geospatial workflows for aggregation, regression modeling, validation, explainability, and reporting using Python, Pandas, GeoPandas, xarray, and SHAP.
- Contributed to research published in the International Journal of Remote Sensing in 2025.

Feb 2023 - Sep 2023

Hyderabad, India

SELECTED PROJECTS

- Enterprise RAG Knowledge System: Built a citation-aware document Q&A; workflow with Python, LangChain, vector search, reranking, FastAPI, PostgreSQL, and Docker.
- Real-Time Streaming Data Pipeline: Built event ingestion and near-real-time processing with Kafka, Spark Structured Streaming, PostgreSQL, FastAPI, and Docker.

SKILLS

Languages: Python, Java, SQL, Bash, JavaScript, TypeScript | Backend: FastAPI, Flask, Spring Boot, REST APIs

Data & AI: Pandas, GeoPandas, xarray, Kafka, Spark, scikit-learn, PyTorch, TensorFlow, SHAP, ETL/ELT

Databases & Delivery: PostgreSQL, MySQL, Oracle, MongoDB, Docker, AWS, Terraform, GitHub Actions, Jenkins, Pytest

EDUCATION

University at Buffalo, SUNY

M.S. Computer Science and Engineering, AI/ML | GPA: 3.8/4.0

Aug 2024 - Jan 2026

Buffalo, NY

Vellore Institute of Technology

B.Tech. Computer Science and Engineering

Jun 2019 - May 2023

India

PUBLICATION

Co-author: "Contribution of Land Use Land Cover Changes to CO2 Emissions Across Koppen Classified Climatic Zones in South Asia (India)," International Journal of Remote Sensing, 2025. doi:[10.1080/01431161.2025.2562005](https://doi.org/10.1080/01431161.2025.2562005)